

# Module End Assignment

## “Analyzing E-Learning Platform Purchases using MySQL”

### 1. Project Overview and Objective:

This project focuses on analyzing the purchasing behavior of learners in an online learning platform. The dataset includes three tables: learners, courses, and purchases, which capture information on learner demographics, course offerings, and purchase transactions. The main objectives of this report are to:

- Understand overall learner spending and purchase trends.
- Identify popular courses and categories.
- Detect patterns in multi-category purchasing behavior.
- Highlight unpurchased courses and revenue contributions per category.

### 2. Problem Statement:

- Identifying which learners spend the most and how spending varies by country.
- Determining total revenue generated per learner.
- Determining the top-selling courses based on quantity purchased.
- Calculating total revenue per course category and the number of unique learners per category.

### 3. Attribute (Column /Features) Details:

|               |         |                                                              |
|---------------|---------|--------------------------------------------------------------|
| Full_name     | VARCHAR | Full name of the learner.                                    |
| Course_name   | VARCHAR | Name of the course.                                          |
| Category      | VARCHAR | Course category (e.g., AI, Data Science, Cybersecurity).     |
| Unit_price    | DECIMAL | Price per unit of the course.                                |
| Purchase_id   | INT     | Unique identifier for each purchase transaction.             |
| Learner_id    | INT     | The learner who made the purchase.                           |
| Course_id     | INT     | The course that was purchased.                               |
| Quantity      | INT     | Number of units of the course purchased in this transaction. |
| Purchase_date | DATE    | Date of the purchase.                                        |
| Country       | VARCHAR | Country of the learner                                       |

#### **4.Descriptive Analytics:**

- Yuki Tanaka spent the most.
- AI and Machine Learning category generated the highest revenue.
- Learners from Japan had the highest average purchase amount.

#### **5. Diagnostic Analytics:**

- Sarah Antony purchased from multiple categories. It shows curiosity or interest in diverse topics.
- AI courses had more unique learners due to high demand or popularity of AI topics.

#### **6. Predictive Analytics:**

- If AI courses show steady growth each month, revenue for the next month can be predicted to rise.
- Learners who purchased multiple high-value courses are likely to spend more on future promotions.

#### **7. Prescriptive Analytics:**

- Recommend AI course bundles for learners who purchased multiple AI courses.
- Target multi-category learners with new Data Science or Cybersecurity courses.
- Offer promotions or discounts on courses that are under-purchased.

#### **8.Key insights:**

- High average spending in Japan suggests targeting premium courses or bundles in that market.
- AI and Machine Learning category generates the highest revenue and has the largest number of unique learners.
- Data Science and Cybersecurity courses have lower purchase counts but still significant revenue, suggesting niche engagement.
- Country-level analysis shows that learners from Japan have the highest average purchase amount, while learners from Saudi Arabia have moderate spending.

#### **9.Conclusion:**

In conclusion, learners' spending patterns reveal high-value individuals like Yuki Tanaka and highlight country-based differences in purchase behavior. All courses have been purchased at least once, indicating good overall engagement, but monitoring low-demand courses can guide future promotions. These insights provide actionable strategies for targeted marketing, personalized recommendations, and optimizing course offerings to maximize revenue and learner satisfaction.

